Organisational Values, Decision Fatigue and Inclusion: Evidence from a Field Experiment

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Abstract

We conduct a field experiment within a business plan competition to examine how institutional features influence inclusive decision-making. In the control group, over the course of independent sequential assessments, evaluators become less likely to recommend female candidates. This pattern reduces the quality of their decisions – as measured by expert assessments and machine-learning methods. Informing judges of the organisation's commitment to equal opportunity entirely offsets this decline, whereas requiring judges to justify decisions to peers has a more muted effect. Our results show that decision fatigue can undermine both decision quality and inclusivity, but simple organisational messaging can counteract this deterioration.

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Recent debates over DEI – diversity, equity and inclusion – have often been framed as arguments about meritocratic evaluation. In Silicon Valley, for example, some firms and investors have advocated replacing DEI initiatives with principles of 'MEI': 'merit, excellence, and intelligence'. Advocates of this alternative view argue that organisations should evaluate candidates solely on quality rather than on demographic criteria: that candidate diversity is often a *substitute* for candidate quality. In contrast, proponents of DEI contend that their efforts correct biases against high-quality candidates from under-represented groups: that candidate diversity and candidate quality can be *complements*.

Within organisations, decisions such as hiring, lending, and grading are typically delegated to individual decision-makers who possess context-specific information but whose biases the organisation cannot fully control (Aghion and Tirole, 1997; Dessein, 2002). These decisions share a common structure: evaluators must make repeated, multidimensional judgments under time pressure, while organisations attempt to align discretion with institutional objectives – for example, by instilling in decision-makers an organisational identity, so that they internalise the organisation's goals as their own (Akerlof and Kranton, 2000, 2005, 2010). This raises a central question (Goldin and Rouse, 2000): can institutions make delegated processes more inclusive without reducing decision quality? This challenge may be particularly acute when decision-makers are busy or fatigued: conditions under which decision quality is likely to deteriorate (Hirshleifer, Levi, Lourie, and Teoh, 2019; Augenblick and Rabin, 2019; Archsmith, Heyes, Neidell, and Sampat, 2025) and implicit biases are likely to surface (Chugh, 2004; Bertrand, Chugh, and Mullainathan, 2005). Following the literature in organisational psychology and behavioural economics, we refer generically to such behaviour as '*decision fatigue*': as evaluators become tired over repeated assessments, they rely more on heuristics and less on careful evaluation (Bodenhausen, 1990; Augenblick and Rabin, 2019).

To study how decision fatigue and institutional design shape both the quality and inclusivity of evaluative decisions, we embed a field experiment in a business plan competition in Ethiopia. In this competition, senior human resources managers from large Ethiopian firms assess young professionals' business proposals for a prize of 50,000 Ethiopian Birr (approximately 800 USD). This experimental design allows us to answer three questions. First, *how does decision fatigue affect inclusive institutional decision-making*? To answer this question, we ask judges to assess repeatedly pairs of candidates in an environment in which other plausible mechanisms are unlikely by design. Second, *to what extent can this be offset by institutional design*? Here we focus on the interaction of two common institutional features: (i) communication of organisational values about equal opportunity, and (ii) the requirement to justify decisions to peers. Specifically, we randomly assign half of our judges to an 'Information' treatment – in which they are exposed to a message emphasizing the importance to the organisation of promoting equal access to capital for female entrepreneurs. We also randomly assign half of our judges to a 'Peer-Justification' treatment, in which they are told in advance that they will be required to discuss and justify their decisions with peer judges – but only after making all their decisions. We randomize these two interventions independently, allowing us to estimate both their individual effects and their interaction.

This leads to a critical third question: *do these institutional features improve decision quality?* In our setting, we use two separate measures to proxy for decision quality. First, we work with two Ethiopian HR experts; we ask these experts to assess each of the candidates, and test for congruence between their assessments and those of our respondent judges. Second, we exploit the fact that our candidates made their pitches by video, recorded in

a studio. This allows us to form factual measures of a large number of characteristics of each candidate (content, tone, facial expressions and appearance); we then use machine learning methods to describe which characteristics are rewarded by judges under different treatments, and how this changes over the course of repeated evaluations. If discrimination is taste-based, and particularly if decision fatigue makes such preferences more likely to shape decisions, then reducing this bias should increase both inclusion and decision quality – by making high-quality proposals from female candidates more likely to be recommended. In contrast, one might imagine treatments making judges more likely to vote for women regardless of proposal quality – in which case inclusivity would improve at the expense of decision quality. The answer matters for organisations: it determines whether institutions face a trade-off between inclusion and decision quality, or whether simple changes to evaluative processes can improve both.

The assessments are structured as a series of binary comparisons – each between a male and a female candidate. Our original design – as reflected in our pre-analysis plan – focussed on average treatment effects over all 12 assessments. This analysis – reported in detail in Appendix A – shows no systematic overall disadvantage for women (and, therefore, no significant average effect of our treatments). This null average result is substantively encouraging in this setting – but also masks sizeable heterogeneity across time. In the control group, evaluators vote for female candidates in 50.0% of the assessments in the first six of twelve assessments, suggesting limited bias against female candidates. In the last six rounds, this falls by 9.9 percentage points, to 40.1%. We show that this decline is driven by worse assessments of those female candidates that, we would have predicted based both on the performance of similar female candidates in early rounds and based on the HR experts assessments, should have won. This indicates a fall in both inclusivity and decision quality.[1] To check that this result is not an artefact of our particular sample, we replicate it drawing on a completely different dataset from a separate experiment with a similar sample of firms.

In the first six rounds, none of our three treatments has a significant effect. However, in the last six rounds, the Information treatment succeeds in eliminating the deterioration of both decision quality and the success rate of female candidates. The Peer-Justification treatment by itself has a directionally similar, though more muted, effect. To measure which female candidates benefit, we exploit our high-dimensional data on respondents' pitches. Specifically, we use two machine learning methods – endogenous stratification (Abadie, Chingos, and West, 2018) and generic machine learning (Chernozhukov, Demirer, Duflo, and Fernández-Val, 2025). Together, these methods show that the Information treatment is particularly valuable for female candidates who would have performed well had they been assessed in the first six rounds (before the decision fatigue took effect). These candidates are significantly more likely – for example – to have a clear business concept and an understanding of the market, are more likely to discuss business operations, are more likely to be perceived as articulate and to use specific examples; their attire is also seen as more appropriate for the competition.

We contribute to two bodies of literature. First, we contribute to the literature on organisational design – by showing how low-cost institutional messages can sustain attention to intended evaluative criteria. Although our experiment is embedded in a specific business-plan competition, the underlying decision environment is common. Evaluators repeatedly make high-stakes, multidimensional judgments under time pressure, while an organisation attempts to align discretion with institutional objectives. Similar problems arise in hiring, promotion, admissions,

[1] Three related papers also study how the sequence of candidates distorts evaluations: Chen, Moskowitz, and Shue (2016); Radbruch and Schiprowski (2025); Kessler, Low, and Shan (2026).

lending, grantmaking, procurement, and entrepreneurial finance. The setting therefore provides a field environment for studying how cognitive constraints affect evaluative discretion, and whether low-cost interventions can restore alignment with intended criteria. A recent literature shows that communicating organisational mission can substitute for financial incentives in motivating effort (Ashraf, Bandiera, and Jack, 2014; Cettolin, Cole, and Dalton, 2024; Khan, 2025). We extend this insight: reminding judges of the organisation's commitment to equal opportunity sustains the effortful processing needed to assess candidates on merit, without constraining their ability to rely on individual expertise. This is light-touch relative to related interventions such as informing evaluators of their implicit biases directly (Alesina, Carlana, La Ferrara, and Pinotti, 2024) or changing evaluation processes (Miller, Lall, Goldstein, and Montalvao, 2023; Bohnet, Van Geen, and Bazerman, 2016), but is also effective. Closer to our treatment, Bursztyn, Fiorin, Gottlieb, and Kanz (2019) show that reminding borrowers of their moral obligation to repay increases repayment – suggesting that simple normative messages can sustain prosocial behaviour. We then show that a second mechanism – anticipation of accountability in front of peers – has a less pronounced effect. Together, our results suggest that low-cost institutional interventions can substantially improve both inclusivity and decision quality by counteracting decision fatigue.

Second, we contribute to a large literature documenting gender disparities in evaluative decisions – including, in particular, in the context of entrepreneurial finance and capital allocation (Ewens, 2023; Hebert, 2025; Alesina, Lotti, and Mistrulli, 2013; Brock and De Haas, 2023; Bartoš, Castro, Czura, and Opitz, 2024). We show that bias against female candidates emerges dynamically over repeated assessments, consistent with decision fatigue.[2] This finding helps reconcile recent evidence from field experiments in Ethiopia showing limited gender bias in access to finance (Ayalew, Manian, and Sheth, 2023; Buehren and Papineni, 2025) with descriptive work highlighting that women face substantially worse access to finance in this setting (World Bank, 2022). More broadly, our results provide evidence consistent with the theoretical distinction between implicit and explicit discrimination (Ridgeway and Correll, 2004; Chugh, 2004; Bertrand et al., 2005) – biases that are absent under full cognitive control surface as decision fatigue sets in (Wigboldus, Sherman, Franzese, and Knippenberg, 2004). Miserocchi (2024) shows that constraints on evaluators' ability to retrieve individual-level information amplify gender discrimination. Bartoš, Bauer, Chytilová, and Matějka (2016) show that discrimination can operate through attention (employers devote less effort to screening minority applicants). Benson, Li, and Shue (2026) find that gender gaps arise from subjective assessments of 'potential' rather than performance.

The remainder of the paper proceeds as follows. Section 1 describes the experimental design, setting and sample in detail, Section 2 presents both the regression and machine learning results, and Section 3 concludes.

1 Experimental design

In this section, we describe the setting for our experiment – a business plan competition in Addis Ababa. We go on to explain the four treatment conditions, and then describe the sample. Appendix C provides detail on the pre-specified experimental design and randomisation.

[2] Kessler, Low, and Sullivan (2019) provide related evidence of this mechanism.

1.1 Setting: A business plan competition

To implement our experiment, we ran a business plan competition among young professionals in Addis Ababa.[3] Competitors were invited to a studio to record a pitch for their business proposal, and these videos form the input for our experiment.[4] The competition took place in a league-style format: each competitor was placed in one out of ten pools of ten candidates, with the winner of each pool receiving a cash prize of 50,000 Birr. Each pool of ten candidates comprised five female candidates and five male candidates. We asked the business plan judges to cast a series of twelve binary votes – where each vote compared one female and one male candidate. For each competition, the candidate with the most weighted votes – pooling judge and HR expert assessments – won the prize.

1.2 Assessing the candidates: Four treatment conditions

The experiment ran over three weeks in the summer of 2022. Randomisation was at the judge level: after agreeing to participate, each judge was assigned to one of four treatment conditions and invited to a session accordingly. Each judge assessed twelve pairs of candidates, and we grouped the assessments so that every pair was viewed by twelve judges — three in each of the four conditions. Both the sequence in which pairs appeared and the order of the female and male videos within each pair were randomised, so that a pair seen first by one judge might (for example) be seen sixth by another; the pair fixed-effects specification therefore compares the same candidates both across treatment arms and when they appeared early rather than late in a judge's sequence. To avoid learning or spillovers, judges were asked not to discuss the candidates with one another until they had completed all of their assessments. We now describe each treatment condition in turn.

Condition 1: Control. Under the 'Control' condition, each judge assessed the videos individually, and without any messaging as to organisational purpose. Specifically, judges were invited to a venue in central Addis Ababa in groups of 12 to 15. All judges entered a single room in a classroom-style arrangement. Once all judges had arrived, the judges were shown two videos. The first video explained the competition and the judges' task (including details on prizes, and on how the competition winners would be determined). In the second video, a prominent Ethiopian businessman and former Olympian, Haile Gebrselassie, spoke briefly about the importance of the business plan competition in providing capital for Ethiopian entrepreneurs. Specifically, he said the following:

> As you know, access to capital is limited for entrepreneurs in Ethiopia. This competition will provide an opportunity for entrepreneurs to access capital to start or grow their business. Your vote is important in deciding which individual will win the 50,000 Ethiopian birr prize; please consider your choices carefully.

[3] Young professionals were eligible for this competition based on their assessment by established firm managers in a separate field experiment: see Abebe, Fafchamps, Koelle, Quinn, and Schwantje (2026).

[4] Specifically, competitors were asked to prepare a three-minute pitch briefly introducing themselves and covering the following five points: (i) their business idea, (ii) their intended target market, (iii) their potential competition, (iv) their operations and (v) their cost of business.

After watching these videos, the judges started their assessments. These were done on tablet computers, with the assistance of facilitators. Specifically, each judge was played a series of video pairs; for each pair, the judge cast a vote for the candidate he or she recommended for the grant. Judges under the control condition knew that their peers were also assessing candidates – but judges were not told anything about which candidates their peers were assessing, and there was no subsequent discussion about any vote cast. Judges were only asked to, privately, provide some feedback to two of the candidates they assessed at the end of their assessments.

Condition 2: Information. The 'Information' condition was designed to emphasise to judges that the organisers of the business plan competition particularly value the inclusion of female candidates. This condition differed from the control condition in just one respect. Specifically, the second video included an additional prompt on the importance of equal opportunity for the organisation – noting, in particular, the specific constraints that female entrepreneurs face in accessing capital in Ethiopia. We deliberately designed this as a light-touch treatment, acknowledging that judges need to consider a range of factors in assessing business plans. Specifically, we inserted into the welcome message the following text:[5]

> Considering equal opportunity: I realise you need to take into account a large number of factors when making your decision but would like you to keep in mind that when starting a business, female entrepreneurs face additional constraints due to lenders' biases. A recent World Bank report finds that male entrepreneurs are more likely to take out loans than female entrepreneurs. In terms of loan sizes, male entrepreneurs borrow about 50 percent more than female entrepreneurs. In this competition, we are committed to gender equality and want to promote male and female entrepreneurs equally.

Condition 3: Peer-Justification. First, before watching the two explanatory videos (and before starting their assessments), judges were asked to introduce themselves to the other judges – providing their name, company and position.[6] Second, prior to each of their binary votes, each judge was shown the names and photographs of two other judges who would also be voting on the same pair of candidates. Third, at the conclusion of all voting, we randomly chose one of the pairwise votes; each judge then sat together with the other judges who assessed that pair (that is, the other two judges whose names and photographs had been shown before that particular vote). In their groups of three, an enumerator shared who each judge voted for, before judges justified their votes to their peers, and then jointly agreed on feedback for the two candidates. Judges under the Peer-Justification condition were told in advance, and before casting each vote, that this meeting would take place.

Condition 4: Information plus Peer-Justification. The final condition combined the features of the Information condition and the Peer-Justification condition: this was designed to test whether communication of organisational values has a different impact when assessors anticipate needing to justify their decisions to peers. Specifically, this

[5] This was inserted after the words 'grow their business' in the Control text. The 'recent World Bank report' referred to is the Ethiopia Gender Diagnostic Report (World Bank, 2022).

[6] The first video was slightly adapted to reflect the change in the protocol.

condition was identical to the Peer-Justification condition, but judges watched the same second video as those in the 'Information' condition.

Theoretical model. To guide intuition in understanding our treatments, we present a game-theoretic model in Appendix B. In this model, evaluators trade off three forces: (i) alignment with their own view of what matters, (ii) organisational pressure to emphasise particular dimensions, and (iii) a cost of disagreeing with other evaluators. The model highlights a simple mechanism. As attention degrades – for example due to fatigue or cognitive load – evaluators place greater weight on their own preferred evaluation rules and reduce alignment with organisational objectives. Organisational messages can partially offset this by re-orienting evaluators toward the intended criteria. These forces generate three predictions that map directly to our experimental design: (i) increases in cognitive costs increase reliance on individual preferences and reduce alignment with organisational criteria, (ii) organisational pressure shifts evaluations without necessarily affecting dispersion, and (iii) greater concern about disagreement compresses variation in evaluations. The model also implies that when alignment is already high, evaluations may already represent the organisational benchmark, so that additional increases in organisational pressure have limited marginal impact.

1.3 Experimental participants: Ethiopian managers

We invited senior Ethiopian managers from established Ethiopian firms to serve as judges.[7] These judges are, by the nature of their roles, deeply familiar with the Ethiopian business environment and the challenges facing firms; they are also a highly relevant sample for testing attitudes towards issues of hiring and inclusion in the Ethiopian labour market. The realised sample contains 245 judges and 2,858 judge–pair observations in the headline estimation sample, including 250 possible unique male–female candidate pairs. The four arms contain 57 Control judges, 59 Information judges, 72 Peer-Justification judges, and 57 Information & Peer-Justification judges. Table A.4 reports summary statistics describing the judges. Most judges are either the most senior manager or owner of the firm (33%) or human resources manager (32%). They are on average 41 years old, and have on average 20 years of professional experience, on average six of which are in their current position. The managers are highly educated, with 78% having a bachelor's degree and 77% having formal management education. Three quarters of the judges are male, one quarter are female. In the final column of Table A.4 we report the p-value for a Wald-test that the judges' characteristics are balanced across treatments, and find no evidence for imbalance for any of the characteristics.

The managers work for large firms with a median of 54 employees (mean 200). 95% of these firms are for-profit, 62% are private limited companies, and 16% are public limited companies. Most firms are located in the capital Addis Ababa, with a few based in the nearby cities of Adama and Bishoftu.

[7] These are drawn from a sample of managers that participated in a separate field experiment: see Abebe, Fafchamps, Koelle, and Quinn (2026). Enumerators are told to closely stick to a script when inviting the judges, and not to give additional information discussing the experiment or how we expect judges to make their decisions.

1.4 Additional assessment: Human resources experts

In addition to the experiment's primary assessments, we had four individuals independently evaluate all competition submissions. This group included two highly experienced enumerators and two senior HR managers. The enumerators focused on transcribing more 'factual' dimensions of the vignettes, such as how the competitors were dressed, how they composed themselves, and whether they addressed each required topic in their pitch. The two senior managers, henceforth referred to as our 'HR Experts', first completed a shortened version of the enumerators' questionnaire. They then assessed the overall quality of the pitch on a scale from one to twenty and evaluated specific dimensions of the pitch, such as whether the competitors had a clear business concept and a strategy for growth.[8] These latter questions were drawn from Fafchamps and Quinn (2018). The experts, who include the owner of an Ethiopian HR firm and the HR head of a large for-profit enterprise, were selected by our local partners for their in-depth understanding of the labour market and ability to assess the quality and viability of business plans.

In our analysis, we interpret the overall score given by these two experts as a proxy for the quality of the proposals. We find that the expert evaluations of male and female candidates are very similar. Among the candidate pairs included in our analysis, female candidates receive scores that are, on average, 0.165 points lower than their male counterparts (on a scale from one to twenty), a difference that is not statistically significant ($p = 0.457$). Similarly, in 47.3% of pairs the female candidate is preferred on average (the "expert favourite"), while in 46.1% pairs the male candidate is preferred ($p = 0.843$); the remaining pairs are tied. Using an alternative, more stringent definition of "unanimous favourite" – where a candidate must be strictly preferred by both experts – the female and male candidate are the unanimous favourite in 26.1% and 28.6% of pairs respectively. Interestingly, the individual experts show more signs of potential biases, with one expert rating the male candidates higher on average, and one expert rating female candidates higher on average. Appendix Table A.5 details these statistics.

2 Results

2.1 Result 1: Decision fatigue in the control group

We begin by documenting gender discrimination in the control group: this pattern is central to motivating and interpreting our subsequent treatment effects. In Table 1, we report three probabilities in the control group: (i) the probability that the female candidate wins, (ii) the probability that a female candidate favoured by the expert wins, and (iii) the probability that a male candidate favoured by the expert wins. We compare each probability between assessments made in the first six rounds and those made in the second six rounds; formally, we do this by estimating the following Linear Probability Model:

$$y_{jp} = \beta_1 \cdot \texttt{Late_Round}_{jp} + \mu_p + \varepsilon_{jp}, \tag{1}$$

[8] We are deliberately agnostic about the criteria: we simply ask the overall score to indicate whether the expert thinks the candidate should be awarded the grant.

where j indexes judges, p indexes pairs, and $\texttt{Late_Round}_{jp}$ is a dummy for whether judge j assessed pair p in rounds 7-12. Because the order in which judges viewed pairs was randomised, `Late_Round` is independent of pair characteristics. The pair fixed effects (μ_p) therefore serve to improve efficiency, by comparing assessments of the same pair of candidates made in early and late rounds.

Column 1 of Table 1 shows that, in the first six rounds, female candidates win 50.0% of pairwise comparisons in the control group. This falls by 9.9 percentage points – to 40.1% – in the second half of the assessments ($p = 0.029$). Column 2 shows that this decline is driven by expert-favoured female candidates becoming 7.0 percentage points less likely to win: this probability falls from 28.4% in the first six rounds to 21.4% in the last six rounds ($p = 0.016$). Column 3 shows that there is no effect for male candidates favoured by the expert (we have a slight positive coefficient of 2.7 percentage points, with $p = 0.393$).

In sum, Table 1 shows that in the second half of their assessments, judges are substantially less likely to vote for female candidates – and, using the expert assessments as proxies for decision quality, that judges are making worse decisions in doing so. Following literature in organisational psychology and behavioural economics, we refer generically to this result as '*decision fatigue*'. This term refers broadly to a range of phenomena by which, as decision makers become more tired, they rely more on heuristic 'rules of thumb' – and such rules often reflect underlying biases (Bodenhausen, 1990; Pignatiello, Martin, and Hickman Jr, 2020). Empirically, this phenomenon has been documented in several settings: financial analysts resort to heuristic forecasts as daily decisions accumulate (Hirshleifer et al., 2019), employers allocate less attention to resumes from negatively stereotyped groups when attention is scarce (Bartoš et al., 2016), judicial rulings deteriorate over the course of court sessions (Danziger, Levav, and Avnaim-Pesso, 2011),[9] and lab participants make more biased decisions in a shooting scenario when cognitively stressed (Ma, Correll, Wittenbrink, Bar-Anan, Sriram, and Nosek, 2013). Archsmith et al. (2025) similarly show that effort spent on earlier high-stakes decisions increases subsequent errors among Major League Baseball umpires.

Indeed, in the same context as the present experiment, Abebe et al. (2026) run an experiment evaluating management traits among young professionals. That experiment does not concern gender bias or organisational values. Nonetheless, the evaluation there required Ethiopian HR managers – similar to the subjects in our experiment – to rank three candidates within each of five triplets of young professionals, assessing their capability as a manager or entrepreneur. In Appendix Table A.13, we show that the result of column 1 of Table 1 replicates in that distinct sample: by the fifth round of assessments in that data, a female candidate is 15.2 percentage points less likely to be ranked first in her triplet, relative to a baseline of 36.5% in round one – a decline of 3.4 percentage points per round in a linear specification.[10]

[9] See also Weinshall-Margel and Shapard (2011) and Glöckner (2016) for alternative explanations for the result in Danziger et al. (2011).

[10] Three of the authors of this paper were authors of that earlier paper: Girum Abebe, Simon Quinn and Tom Schwantje.

2.2 Result 2: The Information treatment offsets this decision fatigue

Motivated by Result 1, we now evaluate the impact of our treatments by expanding equation 1 to interact with our three treatments.[11] Table 2 reports the results, using the same three outcomes as Table 1. It shows that the Information treatment – both on its own and interacted with the Peer-Justification treatment – offsets completely the decision fatigue described earlier. Specifically, the Information treatment increases the probability of a woman winning by 10.5 percentage points in the second half, and similarly by 10.5 percentage points the probability that an expert-favoured woman wins. The equivalent coefficients for the interacted treatment are 5.5 percentage points and 4.8 percentage points (the latter also significant).[12]

In Appendix G we investigate heterogeneity in treatment effects by judges' gender. Although we lack power to detect differences by judge gender due to the small number of female judges, the point estimates suggest that the fatigue-driven decline is concentrated among male judges: they become 10 percentage points less likely to vote for expert-favoured female candidates in the last six rounds in the control group, compared to 4 percentage points for female judges.

2.3 Result 3: The Information treatment improves decision quality

To this point, we have used assessments by local human resource experts as a proxy for decision quality. In Table 1, we found that in the control group agreement with the experts about women fell due to decision fatigue, and in Table 2, we found that this effect was offset by the Information treatment.

To further understand how the Information treatment achieved this, we introduce a large vector of candidate characteristics. Broadly, these covariates fall into three categories. First, we have the human assessments (provided by our human resource experts and by trained enumerators) of proposal content and presentation style. Second, we transcribe and translate the audio using a proprietary algorithm provided by Hasab AI. Third, we use the videos themselves to generate 'emotion AI' measures, using a proprietary algorithm provided by the start-up Affectiva.[13]

To incorporate these candidate characteristics into our analysis, we use two recent complementary machine learning methods. First, we study heterogeneity in treatment effects based on the expected outcome in the control group; to do this, we use the 'endogenous stratification' method of Abadie et al. (2018). Second, we assess which characteristics of candidates predict which female candidates will benefit from the Information treatment in the last six rounds. To do so, we implement the Generic Machine Learning algorithm of Chernozhukov et al. (2025).

[11] Specifically, let $X = \{\text{Information}, \text{Peer-Justification}, \text{Information and Peer-Justification}\}$ be the set of treatment arms (omitting control), let t_j be the treatment assigned to judge j, and denote by $\mathcal{I}$ the indicator function. Then we estimate:

$$y_{jp} = \sum_{x \in X} \beta_x \cdot \mathcal{I}\left(t_j = x\right) + \sum_{x \in X} \gamma_x \cdot \mathcal{I}\left(t_j = x\right) \cdot \texttt{Late_Round}_{jp} + \delta \cdot \texttt{Late_Round}_{jp} + \mu_p + \varepsilon_{jp}.$$

[12] Online Appendix Table A.6 shows this result is robust to interacting treatment with a continuous round variable. The coefficients on the dummy `Late_Round` in Table 2 differ slightly to the equivalent coefficients in Table 1; this is unsurprising, given the way that pair fixed effects enter the estimation.

[13] These include measures of how positive (or negative) the emotions displayed by each candidate are, and how engaged the candidate is during the pitch.

Endogenous Stratification. We implement the method of Abadie et al. (2018) as follows. We repeatedly split the sample, dividing candidate pairs into a main and auxiliary sample. We fit an elastic net model of whether the woman wins using early-round observations from the main sample, and use it to predict outcomes for the auxiliary sample. We then reverse the roles of the two halves, so that no pair contributes to the prediction of how that pair itself will perform in the last six rounds. We repeat this exercise including (i) the full auxiliary sample, (ii) the auxiliary sample assessed in the first six rounds and (iii) the auxiliary sample assessed in the last six rounds. Within each of these samples, we split observations at the median predicted probability and estimate treatment effects separately for the low- and high-predicted-probability groups.

We use these estimates to predict the probability of women winning in the main sample had they been assessed in the first six rounds. We then use a median split on these predicted probabilities – so that we can estimate treatment effects separately for the set of pairs where the woman is less likely to win (the 'low predicted probability' sample) and for the set of pairs where the woman is more likely to win (the 'high predicted probability' sample). Following Abadie et al. (2018), we construct confidence intervals by repeated sample-splitting.

We report results in Table 3. We find clear heterogeneity in the treatment effect of Information in the last six rounds. Women predicted to perform well in the early rounds become substantially more likely to win under treatment (an increase from 50.3 percent to 64.0 percent) – while those predicted to perform poorly exhibit smaller gains (an increase from 36.0 percent to 42.8 percent). We find qualitatively similar patterns of heterogeneity under the interacted treatment, with women predicted to do well in the control group winning 54.2 percent of their assessments, compared to 37.2 percent for those predicted to perform poorly. The Peer-Justification treatment instead does not significantly benefit either subsample, with those predicted to perform well and poorly winning respectively 1.4 and 3.6 percentage points more often. In the first six rounds, none of the treatments significantly benefit any of the subsamples, and we do not find evidence for heterogeneity. We interpret these results as supporting our earlier conclusions: that the Information treatment stops deterioration in the quality of decisions due to decision fatigue.

Generic Machine Learning. The endogenous stratification exercise shows that, in the later rounds, the Information treatment is more effective for candidate pairs in which the woman has a high predicted probability of winning. What characterises these candidate pairs? To answer this question, we implement the generic machine learning framework of Chernozhukov et al. (2025) to study heterogeneity in treatment effects. Appendix F.3 describes the procedure in detail.

In summary, we restrict attention to the last six rounds – where the reduced-form treatment effects are concentrated. To test the impact of the Information treatment, we pool the Information and the Information and Peer-Justification treatment and compare to Control (omitting Peer-Justification for this exercise for simplicity). In each repetition, we split the sample at the pair level into auxiliary and main halves. On the auxiliary half, we estimate separate elastic-net logistic models for treated and control observations using the same rich set of pair-level observables as in the endogenous-stratification exercise. We then use these models to construct – for each observation in the main half – a proxy for the conditional average treatment effect equal to the difference between the predicted outcome under treatment and under control.

We split the sample into two groups based on a median split of the predicted treatment effect. We show that

women indeed benefit more in the group with a high predicted treatment effect: they win 12.1 percentage points more due to the treatment, compared to 5.0 percentage points for the group with a low predicted treatment effect.[14] We can then compare these two groups in terms of their characteristics, we report the results in Figure 1. Strikingly, in pairs with a high predicted treatment effect, experts see the women as having a better business concept, better sense of the market, and a better business sense, and to more frequently mention their business operations. They are also more articulate, give more specific examples, and their attire is seen as more appropriate for the competition by the experts.

3 Discussion

Our results show that inclusive institutional decision-making can deteriorate over the course of an evaluation session, even when there is little evidence of bias at the outset. In the control group, judges were substantially less likely to select female candidates in later assessments, and this decline is concentrated among female candidates who would have performed well had they been assessed earlier. We interpret this pattern as evidence that decision fatigue reduces both inclusivity and decision quality: as evaluators become cognitively depleted, they are more likely to disadvantage strong female candidates. By contrast, a simple prompt emphasising the organisation's commitment to equal opportunity fully offsets this deterioration. The Peer-Justification treatment has directionally similar but more muted effects. Taken together, these findings suggest that low-cost features of organisational design can improve inclusivity and meritocracy simultaneously, not by constraining discretion, but by helping evaluators sustain attention to the organisation's objectives.

To assess whether the treatments affect whether judges identify promising candidates, we collected follow-up data on candidates' labour market outcomes approximately 18 months after the competition. In line with the existing literature on business plan competitions (Fafchamps and Woodruff, 2017; McKenzie and Sansone, 2019), judges' assessments have limited predictive power for candidates' subsequent employment, income, and business investment – and this holds regardless of whether the assessments occurred in early or late rounds. Since only ten candidates won prizes, we estimate these correlations across the full sample, controlling for winning. A caveat is that judges may select candidates whom they expect would benefit most from winning – rather than those with the strongest unconditional prospects – in which case limited predictive power for long-run outcomes may not imply poor selection. We provide additional details in Appendix H.

Our results point to a broader implication for organisations. Much of the public discussion around inclusive decision-making is framed as a trade-off between fairness and performance. In our setting, that framing is misleading. The same institutional feature that improved outcomes for female candidates also improved the quality of decisions. This suggests that some forms of observed exclusion may reflect a failure of institutional design to support careful evaluation under limited attention. Crucially, the classic Beckerian argument that competition erodes discrimination (Becker, 1957; Falco, Gatti, Islam, and Menzel, 2025) applies to taste-based bias, but the fatigue-driven bias we document imposes costs that firms may not recognise – and that competition alone will not correct – without deliberate institutional design. For organisations, our results are encouraging: simple, scalable interventions – such as the

[14] These estimates indicate heterogeneity in treatment effects; formally, the best linear predictor (BLP) test does not reject the null of homogeneous effects ($p = 0.492$).

Information package, perhaps combined with better management of evaluator workload – may improve decision-making without removing discretion or imposing rules. Our findings suggest that understanding discrimination requires attention not only to who decision-makers are, but also to the organisational environments in which they decide. Small changes in those environments can materially affect both whom institutions select and how well they select them.

Table 1: Gender Bias in the Control Group

	(1) Woman Wins	(2) Expert-favoured Woman Wins	(3) Expert-favoured Man Wins
Late Round	-0.099**	-0.070**	0.027
	(0.044)	(0.028)	(0.030)
	[0.029]	[0.016]	[0.393]
Mean (early rounds)	0.500	0.284	0.293
N	651	651	651
Judges	57	57	57
Pair FE	Yes	Yes	Yes

Notes: This table reports OLS estimates of late-round effects on judge decisions using only the control group, regressing each outcome on a Late Round indicator (rounds 7–12). The dependent variables are: (i) woman wins, (ii) expert-favoured woman wins ($= 1$ if the female candidate is both expert-favoured and wins); and (iii) expert-favoured man wins ($= 1$ if the male candidate is both expert-favoured and wins). The omitted category is Early Rounds (rounds 1–6). All specifications include pair fixed effects. We show that these results are robust to including judge fixed effects in Appendix Table A.8. Standard errors clustered at the judge level are reported in parentheses. Wild-cluster bootstrap p-values using judge-level weights are reported in square brackets; significance stars are based on these bootstrap p-values. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 2: The treatment effect in early versus late rounds

	(1) Woman Wins	(2) Expert-favoured Woman Wins	(3) Expert-favoured Man Wins
Late Round	-0.103**	-0.082***	0.026
	(0.042)	(0.027)	(0.029)
	[0.020]	[0.005]	[0.352]
Information × Late Round	0.135**	0.120***	-0.021
	(0.053)	(0.037)	(0.036)
	[0.015]	[0.002]	[0.548]
Peer-Justification × Late Round	0.068	0.071**	-0.004
	(0.051)	(0.034)	(0.035)
	[0.189]	[0.041]	[0.906]
Information & Peer-Justification × Late Round	0.098*	0.084**	-0.014
	(0.055)	(0.037)	(0.038)
	[0.079]	[0.029]	[0.702]
Information	-0.030	-0.015	0.010
	(0.039)	(0.025)	(0.026)
	[0.437]	[0.546]	[0.712]
Peer-Justification	-0.021	-0.024	-0.003
	(0.037)	(0.025)	(0.028)
	[0.588]	[0.326]	[0.907]
Information & Peer-Justification	-0.043	-0.036	0.005
	(0.040)	(0.026)	(0.028)
	[0.279]	[0.169]	[0.874]
Control mean (early rounds)	0.502	0.284	0.294
p: Information (late)	[0.009]	[0.000]	[0.664]
p: Peer-Justification (late)	[0.219]	[0.086]	[0.772]
p: Information & Peer-Justification (late)	[0.193]	[0.097]	[0.715]
N	2,858	2,858	2,858
Judges	245	245	245
Pair FE	Yes	Yes	Yes

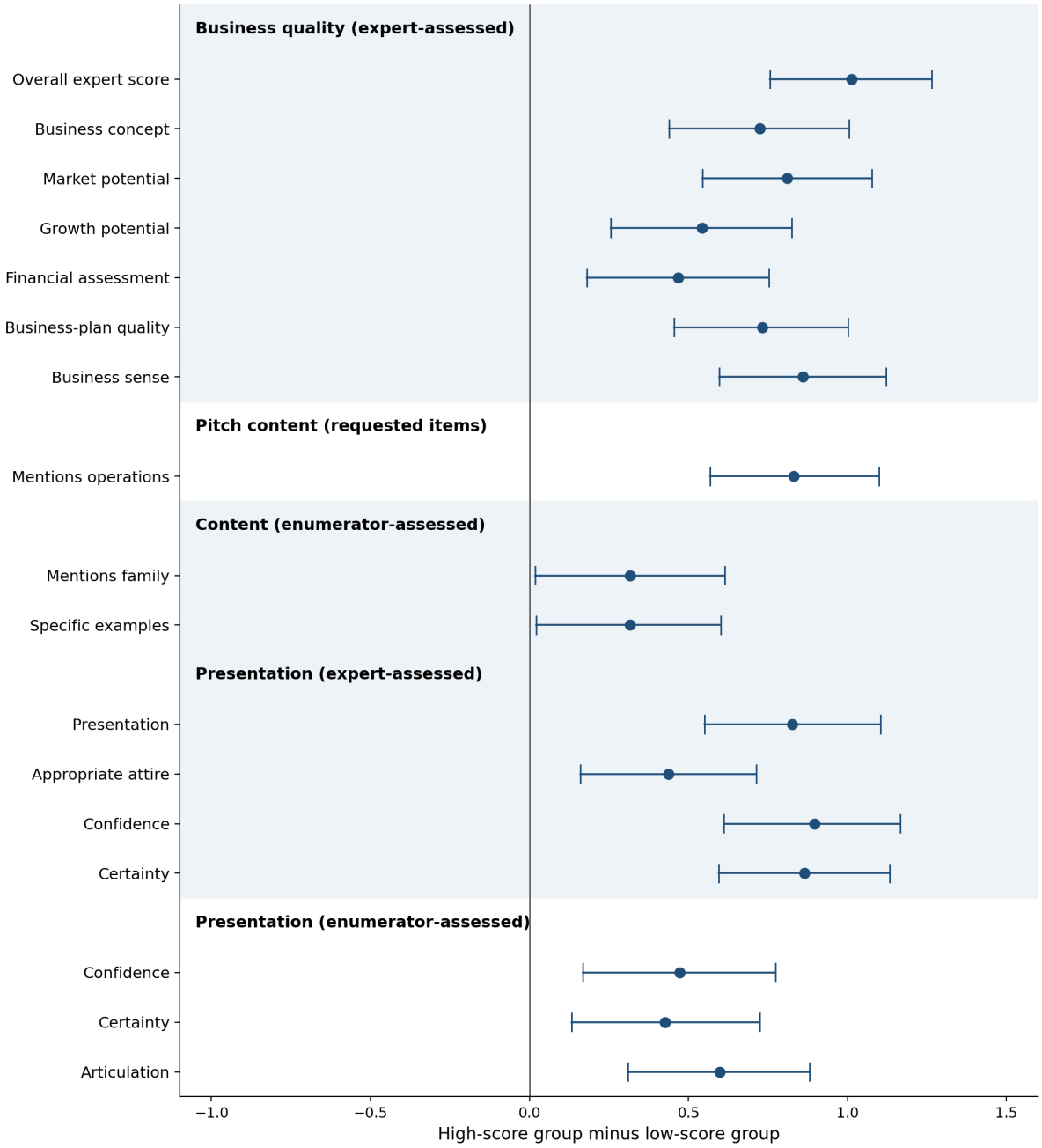
Notes: This table reports OLS estimates of treatment effects on individuals' decision outcomes, interacted with evaluation timing. The dependent variables are: (i) woman wins, (ii) expert-favoured woman wins ($= 1$ if the female candidate is both expert-favoured and wins); and (iii) expert-favoured man wins ($= 1$ if the male candidate is both expert-favoured and wins). Treatment indicators—Information, Peer-Justification, and Information & Peer-Justification—are each interacted with *Late round*, an indicator for rounds 7–12 (late rounds). The omitted category is Control × early rounds. p-values at the bottom test whether the total treatment effect in late rounds (main + interaction) equals zero. "Control mean (early rounds)" reports the mean of the dependent variable for the control group in rounds 1–6. All specifications include pair fixed effects. We show that these results are robust to including judge fixed effects in Appendix Table A.9. Standard errors clustered at the judge level are reported in parentheses. Wild-cluster bootstrap p-values using judge-level weights are reported in square brackets; significance stars are based on these bootstrap p-values. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 3: Heterogeneity in treatment effects on female candidates' performance

	Control	Information	Peer-Justification	Both
Panel A: All rounds				
	Conditional Average Treatment Effect			
Low predicted perf.		1.9% [−3.5, 7.4]	1.2% [−4.3, 6.7]	-1.6% [−7.4, 4.1]
High predicted perf.		5.2% [−0.2, 10.5]	1.7% [−3.5, 6.8]	1.9% [−3.5, 7.2]
	Group Mean by Treatment			
Low predicted perf.	39.5%	41.4%	40.7%	37.9%
High predicted perf.	53.9%	59.1%	55.6%	55.8%
Panel B: Early rounds				
	Conditional Average Treatment Effect			
Low predicted perf.		-2.7% [−10.1, 4.6]	-1.0% [−8.2, 6.3]	-4.3% [−11.6, 3.0]
High predicted perf.		-3.5% [−11.0, 3.9]	1.6% [−5.8, 9.0]	-0.3% [−7.9, 7.3]
	Group Mean by Treatment			
Low predicted perf.	42.9%	40.1%	41.9%	38.5%
High predicted perf.	57.7%	54.2%	59.3%	57.4%
Panel C: Late rounds				
	Conditional Average Treatment Effect			
Low predicted perf.		6.9% [−0.6, 14.3]	3.6% [−3.6, 10.8]	1.2% [−6.8, 9.3]
High predicted perf.		13.7% [6.3, 21.1]	1.4% [−5.3, 8.2]	3.9% [−3.7, 11.4]
	Group Mean by Treatment			
Low predicted perf.	36.0%	42.8%	39.5%	37.2%
High predicted perf.	50.3%	64.0%	51.8%	54.2%

Notes: This table reports results from heterogeneity analysis using the methodology of Abadie et al. (2018). Panel A includes all rounds, Panel B early rounds (1–6), and Panel C late rounds (7–12). Within each panel, the top rows report treatment effects for observations with low and high predicted probabilities of a female win; the bottom rows report outcome levels by treatment arm. Predictions are generated using pair-level sample splitting, so a candidate pair is held out from the model used to score it. We repeat the split 100 times and calculate standard errors and 90% confidence intervals using 1,000 judge-cluster bootstrap replications of the full procedure.

Figure 1: CLAN: Characteristics of Most- and Least-Affected Groups



Notes: This figure reports differences in mean covariate values between the highest-predicted-effect (G2) and lowest-predicted-effect (G1) GATES groups from the Chernozhukov et al. (2025) classification analysis (CLAN). The figure reports covariates with a nominal median-aggregated two-sided p-value below 0.10; Appendix Table A.15 reports all covariates. Each covariate is a standardised female-minus-male difference, so positive values indicate a higher value for the female candidate. The treated group pools treatment arms 2 (Information) and 4 (Information & Peer-Justification) and is compared with treatment arm 1 (Control); treatment arm 3 is excluded. This comparison is therefore not the factorial main effect of Information. The estimation sample is restricted to late rounds (rounds 7–12). Point estimates are medians across 500 repeated pair-level sample splits, and error bars show median-aggregated 90% confidence intervals.

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